# Ranked Execution: Cash-Funded Search for Sparse Portfolio Rebalancing

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**Abstract**

Implementing a target portfolio requires selecting and sizing trades under funding, exposure and minimum-trade constraints. Ranked Execution combines cash-funded marginal priorities, joint convex sizing and bounded search over executable states, including screened trade replacements and restarts from audited alternatives. Mandatory interventions count towards the trade limit; only portfolios passing an original-unit audit can replace the retained incumbent. A matched mixed-integer formulation supplies reference solutions under identical inputs and acceptance rules, with independent and heuristic-started calls reported separately. A portable 302-ETF benchmark supplies 219 monthly decisions with opaque identifiers. Ranked matches heuristic-started MIQP within $10^{-4}$ annual basis points on all decisions, but the ticket allowance is nonbinding. A frozen 24-case synthetic grid compares separate common-base and heuristic-started joint MIQP calls at 10, 60 and 600 seconds. After correcting the reference's numerical scaling and internal feasibility setting, Ranked ties all 24 independent deliveries within the same band at every allowance. Independent MIQP is faster at the median; additional time changes no delivered solution above the band. No reported native bound contradicts an audited witness at the unchanged comparison tolerance. All 182 synthetic arm outputs, including two tiny controls, pass the original audit. Independent enumeration and conservative relaxation bounds supply separate numerical controls. Neither conditional comparisons nor retained-incumbent ties establish global optimality or realised investment performance.

*Keywords:* portfolio rebalancing, trade cardinality, minimum trade, tracking error, mixed-integer quadratic optimization

## 1. Introduction

A model portfolio describes desired exposures. Implementing every change can require many small orders, while selecting only the largest weight changes can overlook their joint risk consequences. The implementation problem is therefore a discrete choice of trades followed by a continuous choice of sizes. Funding, group exposures, minimum trade sizes and minimum held positions couple these choices.

We study a conditional execution problem. Current holdings, target weights, a covariance estimate and execution constraints are supplied inputs. The task is to reduce tracking variance to the target using a limited number of actual noncash trades. A trade limit counts movements from the current book; it is distinct from a limit on terminal holdings. Cash is the settlement coordinate and need not be riskless. Required interventions, such as an exit or an upper-bound correction, consume tickets even when explicitly exempted from a minimum order size.

Ranked Execution evaluates cash-funded marginal directions and then jointly sizes selected executable states. Single-name improvements are supplemented by a bounded shortlist of coordinated state changes. These can admit a funding pair whose isolated trades are unhelpful, or expose another instrument after an earlier useful pair. Every accepted improvement passes the same original-unit audit. A native solver status alone does not establish portfolio acceptance.

The contribution has three parts. First, the formulation keeps original holdings, a prepared feasible base and raw model targets distinct. This separation makes mandatory accounting and minimum-ticket semantics explicit. Second, the search combines inexpensive covariance-aware priorities with joint sizing under the complete coupled constraints. Third, a portable benchmark and independent checks expose failures, search limits, timing and dependence on supplied incumbents.

The empirical input universe comprises 302 publicly traded ETFs. Model weights and covariances are frozen before execution comparisons. Numerical instances have opaque asset identifiers and contain no private fund observations. Readers do not need proprietary signals or a risk-model refit to reproduce execution. Each decision begins from the drifted previous model allocation, so every method receives identical current holdings. This experiment measures conditional implementation decisions, not strategies recursively evolving their own holdings.

A full joint mixed-integer reference fixes the objective, admissible states and return audit. A separate call receives the heuristic's incumbent and charges its generation time explicitly. Three time allowances distinguish additional search opportunity from the effect of that start. Development ablations remain in the replication archive. The separate operational MIQP policy has different objective and constraint options and is outside this matched comparison.

## 2. Related work

Cardinality-constrained portfolio selection and index tracking have a substantial optimization literature. Mutunge and Haugland (2018) analyse covariance tracking with a restricted number of holdings and study greedy construction and exchange improvement. This is a close methodological reference. Here cardinality applies to movements from an existing portfolio, with a distinguished funding coordinate and executable trade/holding intervals.

Restricted-support search is also established. Guastaroba and Speranza (2012) develop Kernel Search for mixed-integer index-tracking formulations by exploring restricted portions of the feasible region. Ranked Execution shares the principle of prioritising small subproblems. Its priority comes from one- and two-coordinate cash-funded quadratics; the subsequent subproblem sizes the entire selected support under coupled constraints.

Strub and Baumann (2018) formulate an index-tracking rebalancing problem that explicitly balances transaction costs and tracking accuracy. Our objective is ex-ante tracking variance with hard operating rules. Transaction-cost minimisation, market impact and execution scheduling are outside the model. A ticket cap is an operating constraint, not a calibrated replacement for every source of trading cost.

Bertsimas and Cory-Wright (2022) develop scalable outer approximation for sparse portfolio selection using ridge regularisation and a convex binary reformulation. This demonstrates the value of strong formulations and carefully designed computation. We do not claim that bounded local search supersedes those methods. Greedy selection, convex sizing and support exchange are established ideas; the present contribution combines them with conditional trade states, funding and audited delivery.

The empirical question is how well this specific combination approximates a matched mixed-integer solution under declared computational controls. Di-

rect comparisons with additional established methods on the same trade domain remain an extension. Differences in objectives and constraints must be resolved before algorithms from other portfolio formulations can serve as matched controls.

## 3. Conditional execution problem

*3.1. Inputs and objective*

Let $c, m, w \in \mathbb{R}^N$ denote current, model and delivered weights, measured as fractions of net asset value. Index $k$ identifies settlement cash and $\Sigma \succeq 0$ is the supplied annual covariance matrix. The objective and reported tracking error are

$$f(w) = (w - m)^\top \Sigma (w - m), \qquad \mathrm{TE}(w) = \sqrt{f(w)}. \tag{1}$$

Annual basis points equal $10^4\,\mathrm{TE}(w)$. No alpha term, turnover penalty or class-specific risk price enters this execution objective.

The long-only hard domain includes

$$\mathbf{1}^\top w = 1, \quad L \le w \le U, \quad a \le Aw \le b. \tag{2}$$

Group rows may be fractional: a mixed-asset instrument need not belong wholly to one group. Bounds and eligibility can encode calendar restrictions and instructions. Cash bounds are real constraints; funding through cash does not imply unlimited borrowing or unrestricted cash deployment.

A terminal noncash holding is zero or at least its declared floor:

$$w_i = 0 \quad \text{or} \quad w_i \ge \theta_i, \qquad i \ne k. \tag{3}$$

An ordinary discretionary movement satisfies

$$w_i = c_i \quad \text{or} \quad |w_i - c_i| \ge \delta_i. \tag{4}$$

The minimum applies to the delivered order, not merely to the desired model-current difference. Joint sizing can otherwise reduce an eligible desired trade to a much smaller actual movement.

Let $\tau_i(w) = \mathbf{1}\{w_i \ne c_i\}$ be the mathematical trade indicator. The total noncash count satisfies

$$\sum_{i \ne k} \tau_i(w) \le K. \tag{5}$$

Numerical delivery uses a separately declared trade-zero threshold. The model does not impose a terminal position-count limit.

*3.2. Mandatory declarations and corridors*

The consumer resolves lifecycle transitions, instructions and ordinary product bounds before risk search. Let $q^0$ be the resulting funded base. Every noncash base-current movement receives a mandatory category. A required correction can have a fixed endpoint or a one-sided interval allowing further movement in the permitted direction. A minimum-ticket exemption removes only the minimum-size condition: it does not remove the trade from the count or waive a holding floor.

For optional instrument $i$, an ordinary corridor lies between its base and target, intersected with eligible hard bounds:

$$[\ell_i, h_i] = [\min(q_i^0, m_i), \max(q_i^0, m_i)] \cap [L_i, U_i]. \tag{6}$$

The actual supplied intervals also retain lifecycle decisions, pins and declared preparation exceptions. Mandatory instruments keep their own resolved intervals. Cash retains its hard funding interval instead of a model-current corridor.

Before solving, declarations must be reconciled with compiled rows. Each enabled group requires an unambiguous label, a finite membership vector and its original lower and upper limits. A missing membership column is an input error, not a zero vector. An intentionally empty group is redundant only if $a_j \le 0 \le b_j$; emptiness must be explicitly acknowledged. A positive lower bound on an empty group is contradictory. Reconciliation records the source declaration, compiled identifier, coefficients and bounds, including distinct identifiers for homonymous groups. The final exposure and residual are reported against those rows. This pre-solve check closes a gap that a return audit alone cannot detect: an omitted constraint is absent from the audit as well. Reconciliation changes no objective coefficient, endpoint or numerical tolerance.

The eligible hard domain permits movement within full resolved product and instruction bounds. We call the two domains CORRIDOR and FULL. They answer different optimization questions. Every algorithm in the main comparison uses CORRIDOR. FULL is used by common preparation only when the ordinary corridor cannot supply a valid base.

If the policy declares a discretionary allowance $B$, the total cap is $K = M + B$, with $M$ unique active mandatory names after preparation. This is an explicit convention. An input declaring a fixed total cap must retain it; preparation cannot silently enlarge that cap when declaring another repair.

*3.3. Common feasibility preparation*

Individual lifecycle decisions can produce a base that violates funding, group bands or holding floors. Separately repairing each algorithm's base would conflate preparation and risk-search effects. We prepare one shared instance before running any comparison arm.

An already audited base is retained exactly. Otherwise, preparation searches CORRIDOR and then, if needed, eligible FULL bounds. Its objectives are risk-neutral. In FULL, corridor excursion is minimised before gross noncash intervention and the number of additional repair instruments. Within the selected discrete pattern, a strictly convex squared-intervention objective breaks continuous ties. Target risk and alpha do not choose among repairs.

Additional noncash interventions are pinned, declared and counted before risk search. Endpoints outside an original corridor are recorded as preparation exceptions, while full hard bounds and eligibility remain unchanged. All arms receive the same final base $q$, intervals, mandatory flags, exemptions and resolved cap.

Sequential optimization stages use an explicit numerical objective guard; they are not exact symbolic lexicographic arithmetic. A shared 30-second allowance includes preparation stages. Failed preparation remains in the experiment denominator. A numerical failure or exhausted budget alone does not establish mathematical infeasibility. Conditional solver comparisons state how many inputs successfully prepared.

## 4. Executable states and matched formulation

*4.1. Finite unions of intervals*

For each noncash instrument $i$, compile a finite family of executable alternatives, $\mathcal{J}_i = \{[\ell_{ij}, h_{ij}] : j \in J_i\}$. For an ordinary optional instrument these include the no-trade point when valid, a sell interval ending at $c_i - \delta_i$, and a buy interval starting at $c_i + \delta_i$. Intersect the alternatives with the permitted corridor and with $\{0\} \cup [\theta_i, \infty)$, and remove empty intervals. Mandatory exemptions and fixed pins are applied before compiling the states.

Each state carries a ticket flag $t_{ij} \in \{0, 1\}$ measured from original holdings. An optional no-trade point has flag zero; an exempt mandatory movement keeps flag one. Once states are fixed, remaining sizing is a continuous convex problem.

For example, let $c_i = 0.02$, the corridor be $[0, 0.04]$, $\delta_i = 0.005$ and $\theta_i = 0.01$. The alternatives are a no-trade point at 0.02, a full exit at zero, a sell interval $[0.01, 0.015]$, and a buy interval $[0.025, 0.04]$. Selecting a sale does not prescribe its unique size. Funding and group constraints determine final size jointly with other names. Several states never create several tickets for one instrument.

*4.2. Full joint MIQP reference*

Introduce binary selectors $z_{ij} \in \{0, 1\}$ and continuous contributions $v_{ij}$. The full joint model chooses every state and trade size together:

$$\min_{w,v,z} \quad (w - m)^\top \Sigma (w - m) \tag{7}$$

$$\text{subject to} \quad \sum_{j \in J_i} z_{ij} = 1, \qquad \ell_{ij} z_{ij} \le v_{ij} \le h_{ij} z_{ij}, \tag{8}$$

$$w_i = \sum_{j \in J_i} v_{ij}, \qquad \sum_{i \neq k} \sum_{j \in J_i} t_{ij} z_{ij} \le K. \tag{9}$$

and all funding, instrument and group rows in (2). This is a convex mixed-integer quadratic problem (MIQP). A declared optional allowance adds its corresponding sum over optional names. Cash has no trade selector.

Every method uses the same prepared state domain. "Full joint" means all choices are optimised together; it does not mean replacing the prepared CORRIDOR domain with the wider FULL preparation domain. The model's optimum cannot be worse than a feasible heuristic vector in this domain. The historical replication identifier `Exact` names the formulation, not a certificate for every returned vector. The implementation solves an equivalent conic canonical representation. Native status, native incumbents and externally retained weights remain separately attributed.

The operational `execution/MIQP` implementation is a separate policy: it permits an alpha trade-off, a terminal position cap and repair-on-touch rules. Those options are absent from this tracking-variance benchmark. Its outputs would confound policy and algorithm effects, so they are not included as an additional matched arm.

Fixing one state per instrument yields a convex quadratic program. Fixed coordinates and cash can be eliminated algebraically. Heuristic supports use the same joint-sizing formulation. A conditional fixed-state refinement of an integer incumbent preserves its declared state choices and is audited separately; native outputs are never rounded into acceptance.

## 5. Ranked Execution

### *5.1. Cash-funded marginal gains*

A change in noncash instrument $i$ is funded through $u_i = e_i - e_k$. At working point $w$, define

$$g_i = u_i^\top \Sigma(w - m), \qquad s_i = u_i^\top \Sigma u_i \geq 0. \tag{10}$$

A displacement $x$ changes risk by

$$f(w + xu_i) - f(w) = 2g_i x + s_i x^2. \tag{11}$$

For state displacement interval $D_i = [d_i^-, d_i^+]$, minimise this scalar quadratic. When $s_i > 0$, the minimiser is the projection of $-g_i/s_i$ onto $D_i$; when $s_i = 0$, an endpoint minimises the linear expression. Ties are resolved deterministically.

The priority is the negative objective change at that minimiser. Greedy selection admits positive-gain optional names within the allowance and updates covariance products after each virtual move:

$$\Sigma(w - m) \leftarrow \Sigma(w - m) + x_i(\Sigma_{:i} - \Sigma_{:k}). \tag{12}$$

A selected name cannot consume another ticket by choosing a second state. Instrument identifiers and state indices break score ties.

Virtual working points are support proposals. They need not satisfy coupled cash or group bounds. Those constraints enter joint sizing, and only an audited jointly sized vector can replace the incumbent. A marginal priority guides computation without becoming an acceptance rule.

### *5.2. Joint sizing and mandatory coordinates*

The search starts from the shared audited incumbent $q$. A mandatory-only QP first explores movable mandatory coordinates and their bounded holding alternatives. A feasible base can still be improved along these coordinates; an empty optional support must not suppress their optimization.

For a proposed optional support, selected states are free within their intervals; unselected optionals remain at base. Movable mandatory coordinates and cash are jointly sized with them. Funding, instrument bounds and original group rows are included. Every candidate is audited before retention.

The default explores at most 32 mandatory state combinations per support. Truncation is reported. Failure of one assignment does not prove that

every assignment for those instrument names is infeasible. A feasible candidate with more risk than the incumbent leaves the incumbent unchanged.

If initial greedy sizing supplies no audited candidate, a bounded rescue list tries constraint-directed admissions, removals and replacements. Its separate allowance is at most 24 support trials. Rescue respects the same cap and states; it cannot introduce a new exemption or broaden a corridor.

*5.3. Coordinated state changes*

Isolated priorities can miss a useful combination. Funding and group restrictions may make a buy and sale jointly attractive even when their isolated proposals are unsuitable. The method supplements one drop and one exchange pass with a bounded coordinated neighbourhood.

At incumbent $\bar{w}$, let $C_{ij} = u_i^\top \Sigma u_j$. For proposed state-displacement intervals $D_i, D_j$, the two-coordinate priority is

$$-\min_{x \in D_i,\, y \in D_j} \{2g_i x + 2g_j y + C_{ii}x^2 + 2C_{ij}xy + C_{jj}y^2\}. \tag{13}$$

The positive-semidefinite box quadratic is evaluated through boundary minima and any admissible interior stationary point. This priority is inexpensive relative to complete joint sizing.

The neighbourhood changes one or two optional states: admissions, removals, replacements within available slots, or holding alternatives. Priorities do not include all coupled rows or reoptimisation of the other selected coordinates. Consequently, even a negative isolated priority need not exclude a support from the shortlist.

The best unvisited candidates form a bounded shortlist. Joint sizing evaluates them under all constraints. If the incumbent improves and trials remain, the search restarts from the new incumbent. At most three anchors share a total coordinated budget of 24 support trials. A pass without improvement stops the stage. These are support trials; mandatory alternatives and numerical retries can require more than 24 native QP calls.

*5.4. Scarce-ticket extension*

When the optional candidate count exceeds a positive allowance, the completed search above is retained and a separate branch explores scarce-ticket combinations. First, solve the continuous perspective model of the same state formulation. In funded coordinates $w = r + Ux$, write $H = U^\top \Sigma U$ and choose $D = \mathrm{diag}(d)$ with $d \geq 0$ and $H - D \succeq 0$. The implementation

uses half the nonnegative minimum eigenvalue of the diagonally normalised active block of $H$, multiplied by its diagonal, and verifies the residual matrix. Retain the constant and affine terms of $f(r + Ux)$ and replace its quadratic part by

$$x^\top (H - D)x + \sum_i d_i x_i^2 / z_i,$$

where $z_i$ is the sum of optional ticket-state selectors and equals one for mandatory coordinates. The closed perspective uses zero at $(x_i, z_i) = (0, 0)$ and excludes $x_i \neq 0$ when $z_i = 0$. Relax selectors to their continuous hull only for this guidance solve. At executable integer states the objective is the original quadratic; the integer reference remains the matched formulation above. Rank optional names by fractional ticket occupancy, choose each name's highest occupancy ticket state, and round the first $B$ names into a support. Identifiers and state indices resolve ties. Fractional weights are never returned: the rounded support must pass the unchanged joint-sizing QP and original-unit audit. An audited seed may initially be inferior; its search branch is kept separate and the original incumbent is restored unless the completed branch strictly improves it.

At each scarcity anchor, jointly size every one-for-one optional replacement. This sweep allows all retained coordinates to resize. Merely scoring the removed and admitted coordinates can miss an improvement that requires that recourse. The expanded neighbourhood also admits two new names coupled to one or two removals, making a two-for-two replacement reachable at a full ticket budget. Its two-coordinate priority includes cash and group rows through polygon clipping and quadratic minimisation on edges and the interior. If that fixed-coordinate slice is empty, retain the box priority in (13): other selected names can move during joint sizing, so an empty slice is not a support infeasibility proof. For a coupled removal, include its exact cash-funded quadratic change in the score.

The rounded seed and expanded shortlist share a separate 24-support allowance. Up to three anchors each have a complete single-replacement sweep; these sweeps are additional work, not part of the 24-support count. After a shortlisted improvement, restart early when another anchor is available. Stop after an unimproved pass. Search is bounded but neither local nor global optimality follows. The extension uses no integer solve and no reference-arm incumbent. Its continuous model supplies priorities only; it does not change the delivered objective, constraints, mandatory declarations or acceptance

tolerances.

The final pair replacement can expose a useful single replacement after the last sweep. Also, an inferior or infeasible rounded seed can lead to a better support through joint resizing. We therefore run a final replacement branch from the completed incumbent and from the unrestricted rounded seed. To diversify beyond pair replacements, pin one selected name at its no-ticket state in a temporary continuous guidance model, round the resulting occupancies and search that seed. At most $\min(B, 24)$ incumbent names are tried, in identifier order; duplicate seeds are skipped. The pin affects guidance only: every candidate is sized and audited on the original problem, and the excluded name may re-enter a branch.

Each distinct seed receives at most three complete single-replacement sweeps. If its initial sizing fails, the first sweep still tries replacements around the proposed support. Subsequent sweeps use the best audited branch support and stop without improvement. Each branch is compared to the separately retained completed incumbent. A per-call cache stores the best audited original-domain sizing result, or failure, for each support over the same bounded mandatory assignments. Separate branch visitation allows a cached support to serve as an intermediate point without another QP. This bounded diversification can cross more than two replacements in total; it is not exhaustive multi-name enumeration.

An empty optional incumbent has no names to exclude. Moreover, a promising combination can require retaining a particular trade, rather than removing one. A further bounded phase therefore takes the union of the completed incumbent's ticket states and the unrestricted rounded seed's ticket states. In stable instrument/state order, it conditions on inclusion and then exclusion of each state, stopping after at most 24 additional continuous solves. Inclusion fixes that state selector to one; exclusion fixes it to zero. All other objective terms, rows and numerical tolerances remain unchanged. The rounding step respects the decision, prioritising a forced admission over unconstrained choices.

Each distinct proposal receives the same replacement restart, including rescue around infeasible seeds. A final restart revisits the completed incumbent. These temporary selector conditions guide proposals only: executable sizing uses the original problem, and every branch competes with the retained audited incumbent. This phase is a finite conditional neighbourhood, not recursive branch-and-bound. Its continuous objective values are not optimality certificates.

*Screened pair completion.* Slice priorities hold retained positions fixed, whereas joint sizing can resize them. Consequently a low-priority pair can still improve the portfolio. An additional phase shares at most $R^2$ additional coordinated trials across the same anchor limit, where $R = 24$ is the rescue allowance: 576 trials and at most three anchors by default. Each anchor retains the complete single-replacement sweep. All previously sized supports are skipped; strict audited improvements re-anchor the search while passes remain. The earlier incumbent is always retained.

An inexpensive necessary-feasibility screen precedes the shortlist. Selected coordinates and cash span their original intervals, unselected coordinates stay at the base, and mandatory coordinates span the union envelope of their states. For each funding or group row, interval arithmetic gives its smallest and largest possible value, using the coefficient sign to select each endpoint. Existing coordinate and row audit allowances yield a conservative outer envelope. A support is rejected only if this envelope cannot intersect a required row interval. Surviving supports still pass unchanged original-domain joint sizing and audit; the screen neither relaxes an executable constraint nor proves joint feasibility. It removes obvious dead ends without suppressing recourse in retained positions. The finite trial and anchor budgets can still miss improving combinations.

*Cached alternative restarts.* Improving-only descent can stop before a feasible three-position replacement. The sizing cache already contains audited alternatives that were not retained. Order these by objective and then instrument/state identifiers, exclude the current support, and restart from at most $R$ alternatives. Each branch uses the same bounded complete single-replacement sweeps and separately retains the incumbent. A worse seed may therefore connect a useful sequence of supports. At the first branch improving the incumbent, complete its pair neighbourhood once using another $R^2$ trial allowance and the same anchor limit, then stop. The support list is selected before these restarts; it uses no reference-arm portfolio. Failed branches cannot worsen delivery. This finite search broadens coverage without establishing exhaustive three-position optimality.

### *5.5. Properties and computational cost*

**Proposition 1 (Conditional incumbent retention).**

*For fixed arrays and acceptance rules, if the initial incumbent passes the*

Table 1: Ranked Execution with audited incumbent retention.

| Step | Operation |
|---|---|
| 1 | Validate inputs, compile states and audit the shared prepared base. |
| 2 | Size movable mandatory states without optional trades. |
| 3 | Construct cash-funded greedy priorities and jointly size selected states. |
| 4 | If needed, attempt the bounded feasibility-rescue shortlist. |
| 5 | Attempt one drop and one exchange pass over the initial accepted support. |
| 6 | Jointly size coordinated changes; restart after improvement while the shared trial and anchor allowances permit. |
| 7 | Under ticket scarcity, retain this incumbent; jointly size a perspective-guided seed, complete single replacements and shortlisted pair replacements. |
| 8 | Search final replacement sweeps and distinct seeds from temporarily excluded incumbent names; size every candidate on the original problem. |
| 9 | Include/exclude states from the incumbent and tentative relaxed seed in bounded guidance calls; search distinct proposals and revisit the completed incumbent. |
| 10 | Revisit the completed incumbent with the interval-screened pair allowance; jointly size each shortlisted support on the original domain. |
| 11 | Restart from bounded cached audited alternatives; after the first improving branch, complete its pair neighbourhood once. |
| 12 | Return the best audited branch, with all search work and failures recorded. |

*return audit and replacement requires an audited strict objective improvement, the returned objective is no worse than that of the initial incumbent.*

PROOF. The initial incumbent is stored. Each replacement passes the same audit and has smaller objective. Failure to find a replacement leaves it unchanged.

This delivery property is equally available to the mixed-integer reference. It is not an approximation ratio or an optimality certificate and says nothing about future risk after rolling holdings diverge.

For dense covariance, the initial covariance-vector product costs $O(N^2)$. With bounded states per instrument, each greedy score and column update

costs $O(N)$, giving $O(N^2 + BN)$ arithmetic for that stage and $O(N^2)$ input storage. These counts exclude validation, convex sizing, mandatory enumeration and improvement search.

With $s$ available state changes, coordinated enumeration can inspect $O(s^2)$ pairs per anchor. A bounded shortlist does not make its generation constant work in universe size. Support-key construction and ordering add overhead. The scarcity extension can enumerate $O(B^2 s^2)$ pair replacements per anchor, plus at most $Bs$ jointly sized single replacements. It also solves one unrestricted and up to $\min(B, 24)$ restricted continuous perspective problems. The final incumbent, unrestricted seed and distinct restricted seeds add at most $2 + \min(B, 24)$ branches, each with at most three sweeps of $Bs$ supports. If the conditional phase considers $C \leq 24$ decisions, it adds $C$ continuous solves and at most $C + 1$ replacement branches. Each branch sizes its seed and at most three sweeps of $Bs$ supports. Caching removes duplicate sizing work within a call. Each earlier stage has at most three anchors and its own shortlist allowance; mandatory enumeration multiplies sizing counts. The screened completion phase adds at most $R^2$ coordinated supports and three single-replacement sweeps. Its interval screen costs $O(mn)$ per candidate for $m$ group rows and $n$ instruments, in addition to pair-priority generation. Cached alternatives add at most $R$ branches of three $Bs$ replacement sweeps. The first improving branch permits at most one further completion phase with the same bounds. Cached results avoid duplicate QPs, but scoring and revisiting supports still incur work. These additional costs are measured rather than hidden inside the initial greedy complexity. A support cap is not a wall-clock guarantee.

The library accepts exact factor-plus-diagonal covariance representations when available. The empirical comparison retains its supplied dense matrices, including residual correlations. Substituting a diagonal residual approximation would change the experiment and is not done here.

## 6. Numerical acceptance and comparison protocol

### *6.1. Original-unit audit*

Every final vector is checked in NAV fractions against unchanged supplied arrays. Feasibility tolerance is $10^{-7}$ and actual-trade zero is $10^{-8}$. Holding floors and minimum tickets retain their declared values. The independent benchmark audit checks funding, instrument intervals, group exposures, floors, nonexempt ticket sizes and both ticket allowances. It calls neither the

implementation's state compiler nor its audit evaluator. Returned vectors are not renormalised or rounded.

Numerical QPs express movable weights in basis points, eliminating fixed coordinates and cash algebraically. In fixed-state sizing, a bound or group row is removed only when every remaining variable coefficient is exactly zero and its constant value satisfies both unchanged endpoints. Nonzero coefficients, however small, and constant contradictions are retained. This algebraic reduction avoids degenerate constant rows without enlarging the feasible set. Integer hulls retain every linear row; their quadratic cone is balanced as described below. Continuous relaxation graphs are unchanged. Group rows use the same bounded residual envelope as the original-unit audit. Original limits remain authoritative at return. CLARABEL uses absolute/relative gap and feasibility controls of $10^{-10}$. A numerical solver exception permits one fresh attempt with equilibration disabled, retaining the graph, objective, rows and tolerances. An infeasible status does not trigger that retry.

The qualified implementation reuses compiled sizing graphs only when canonical sparse coefficients agree exactly with a fresh construction. Each trial still receives a fresh numerical solve and the same audit; no trials or constraints are dropped. A failed equivalence check uses the ordinary construction. This reduces compilation overhead without changing the search or measured weights.

This aligns numerical construction and acceptance. It does not prove literal feasibility beyond the declared tolerance, or convert a native lower bound into a certified optimality gap. The benchmark records raw statuses and candidate rejection alongside delivered portfolios.

### *6.2. Three principal comparisons*

Table 2 specifies the principal comparisons. Common preparation is performed once and fixed across arms. Common-base MIQP starts from the prepared incumbent. V2-started MIQP is a separate call supplied with Ranked's incumbent. Neither call receives a pooled best result across algorithms.

The ETF comparison uses a 10-second wrapper allowance; the synthetic study uses separately started 10/60/600-second calls. All use one native thread, seed zero and relative MIP-gap setting $10^{-4}$. Graph construction, start lifting and conditional fixed-state refinement consume this allowance. Native operations can overrun it; measured wall times are reported.

Table 2: Arms under the same prepared CORRIDOR domain.

| Arm | Definition |
|---|---|
| Ranked (V2) | Full default search, including all scarcity extensions; up to three anchors per stage and separately recorded sweeps. |
| MIP-B | Full joint model started from the prepared base (`ExactBase`). |
| MIP-V | Separate full joint solve started from V2 (`ExactRanked`). |

An earlier native integer feasibility setting of $10^{-4}$ produced a false optimal termination on the overlapping-group instance `joint_007`. The native lift of a better feasible portfolio reproduced its objective, and an independently formulated full-weight fixed-support QP confirmed the witness. Tightening only native integer feasibility to $10^{-9}$ recovered the better solution without supplying it as a start. Disabling presolve also recovered it; additional time alone did not. Tightening that setting alone left smaller bound inconsistencies. For integer hulls we therefore also replace $x^\top Qx/10^4$ by $(x/100)^\top Q(x/100)$ in the conic graph, where $x$ is funded displacement in basis points. This balances the quadratic epigraph without changing its value, any linear row or the native start mapping. Fixed-state sizing and continuous relaxation graphs are unchanged. The correction changes internal numerical accuracy, not the objective, state constraints, original-unit audit or other solver tolerances. The ETF and full joint tables use both corrections; earlier search-development records retain their original source hashes.

Raw native statuses and bounds remain visible. A consistency check compares a native bound with the delivered incumbent, and the benchmark separately checks it against every audited cross-arm witness at the existing $10^{-12}$ variance comparison scale. Bounds are never clipped to an incumbent. Passing these checks is not an independent global certificate.

The ETF replication also retains Greedy (without support rescue or improvement) and OneAnchor (one coordinated anchor per stage and one replacement sweep per diversification branch). Their results and the earlier search-development studies remain available as ablations rather than additional headline methods.

Native audited incumbents, retained external starts and refined candidates have distinct attribution. An ExactRanked delivery equal to Ranked does not establish independent native discovery. An ExactBase delivery worse than Ranked under a time limit does not mean its mathematical opti-

mum is worse.

Arms run serially on one host. Algorithm-call times exclude imports and input loading. Preparation is reported separately and included in end-to-end time. ExactRanked additionally includes its Ranked start-generation cost. These are declared computational controls, not equal elapsed budgets. The recorded host runs Windows 11 on an Intel Core Ultra 7 265 processor. The rerun uses Python 3.12.14, unreleased corrected OP source with 9.1.5 metadata, QIS 5.44.4, CVXPY 1.9.2, MOSEK 11.2.3 and CLARABEL 0.11.1. The public PyPI 9.1.5 wheel does not contain this correction; accompanying source hashes identify the implementation. The replication protocol records the full numerical package versions and execution-source hashes.

An earlier archived ETF run used a different preparation implementation and dependency cohort; its allocation changes cannot be attributed to this search extension alone. A separate regression control repeats all 219 decisions with published OP 9.1.5 under the current dependencies. Its Ranked weights and TE values are exactly identical to the corrected source on every decision. The scarce-ticket extension is inactive in these cases. Control timings are excluded because verification ran concurrently; the five-arm timings below come from the serial primary run.

## 7. The 302-ETF benchmark

### *7.1. Frozen supplied inputs*

The source universe contains 302 publicly traded ETFs, retrospectively selected in October 2026. Frozen model allocations span June 2008 through September 2026. We construct 219 monthly decisions from July 2008 onward. Eligible covariance support expands with available observations. The supplied universe is not a point-in-time investable-universe reconstruction.

Each case begins from the previous supplied model portfolio, normalised once to unit investment and drifted through the intervening month with the canonical portfolio helper. This removes the previous model's small investment residual when defining the hypothetical current book. The current raw target remains unchanged in the risk objective. No competing execution strategy's historical holdings define these cases.

The experiment is conditional on supplied targets and risk estimates. It does not validate upstream model selection, estimate the economic value of signals or establish out-of-sample investment performance. Monthly cases share assets and market history and are not treated as independent samples.

Every instrument has a 10% upper bound, including settlement cash; unavailable instruments have zero upper bound. Exposure bands are 0–10% liquidity, 25–45% fixed income, 35–65% equity and 5–15% alternatives. A mixed-allocation instrument has fractional equity/fixed-income loadings. Monthly eligibility and lifecycle declarations precede common preparation.

Ordinary actual tickets are at least 25 basis points of NAV. Equity and alternative holdings have a 50-basis-point floor; fixed income and liquidity have no positive floor. The optional allowance is ten tickets plus the frozen mandatory reservation. Compulsory interventions retain their minimum-ticket exemption and remain counted. There is no separate position-count cap.

### *7.2. Public replication boundary*

The inputs contain current and target weights, the pre-preparation base, annual covariance, corridor and full bounds, group rows, eligibility and fixed-coordinate flags, mandatory categories, exemptions and ticket rules. Stable identifiers such as `Asset001` replace instrument labels; `Cash` identifies funding.

There are no raw vendor price histories, private portfolio observations, factor-estimation inputs, local filesystem paths or private identifier mapping in the replication dataset. The supplied arrays reproduce common preparation and all execution arms. Model targets alone would not suffice.

Opaque identifiers provide de-identification, not a guarantee that numerical histories cannot be recognised. The benchmark concerns publicly traded instruments and makes no claim to conceal their market characteristics. Stable labels and unchanged coordinate order avoid conflating renaming with algorithm changes.

Every data file is hashed before loading. Arrays contain no Python pickle objects. Reference outputs are used only for verification and reporting, never to select a support or improve an incumbent. Replication calls reusable OP execution and QIS risk analytics without a private application or an upstream research repository.

### *7.3. Results and interpretation*

All 219 decisions were attempted. Common preparation succeeded on 219; all five arms supplied independently audited portfolios on 219 decisions. The archive retains every attempted case, including any failed preparation. Comparisons use paired audited outputs.

Table 3 uses the common audited subset. Tracking error is annual basis points; computation is elapsed seconds. The prepared base has mean tracking error 32.897 basis points. Median common preparation takes 0.076 seconds.

Table 3: Conditional execution quality and measured computation. End-to-end (E2E) includes common preparation and, for ExactRanked, Ranked start generation.

| Arm | Mean TE | Median TE | Median s | 95th pct. s | Median E2E s |
|---|---|---|---|---|---|
| Ranked | 16.975 | 4.349 | 0.061 | 0.257 | 0.192 |
| ExactBase | 16.975 | 4.349 | 0.056 | 0.143 | 0.140 |
| ExactRanked | 16.975 | 4.349 | 0.055 | 0.081 | 0.277 |

Pairwise differences use a descriptive equality band of $10^{-4}$ basis points, distinct from the portfolio feasibility tolerance. Negative differences favour Ranked. No independence-based significance test is applied to these serially related decisions.

Table 4: Ranked relative to each comparator on paired audited decisions. The last column is mean Ranked minus comparator TE, in basis points.

| Comparator | Pairs | Ranked lower | Equal | Ranked higher | Mean difference |
|---|---|---|---|---|---|
| ExactBase | 219 | 12 | 207 | 0 | -0.00002 |
| ExactRanked | 219 | 0 | 219 | 0 | 0.00000 |

Optional candidate counts range from 0 to 5 (median 1); 107 prepared cases have no optional candidate. Mandatory reservations range from 8 to 70 (median 30). These are the effective search dimensions, not the nominal 302-asset universe.

Preparation retains 65 already audited bases, repairs 79 within the ordinary corridor, and uses FULL preparation on 75 decisions. The resulting pins and corridor exceptions are shared by every arm.

The optional ten-ticket allowance never binds in this sample: every prepared case has at most five optional candidates. Mandatory interventions dominate the ticket count. Thus these data do not test scarce-ticket selection among a large optional candidate set.

Ranked improves on Greedy in 10 cases and is equal in 209. OneAnchor and the full search have equal delivered objectives on 219 paired cases. ExactBase median call time is 0.056 seconds versus Ranked's 0.061. The measured computation does not support a general heuristic speed advantage.

ExactBase delivered-source attribution is *common prepared incumbent*: 65, *native callback*: 140, *native return*: 14. It recorded a native audited candidate on 219 of 219 prepared cases. A callback or native return may reproduce a submitted start; this count alone is not evidence of an independently discovered improvement.

ExactRanked delivered-source attribution is *common prepared incumbent*: 65, *native callback*: 55, *native return*: 7, *requested mip start*: 92. It recorded a native audited candidate on 219 of 219 prepared cases. A callback or native return may reproduce a submitted start; this count alone is not evidence of an independently discovered improvement.

The largest delivered TE disadvantage of Ranked against ExactRanked is 0.00004 basis points (case 0046). Delivered differences compare feasible incumbents, not certified global gaps. Full weights, numerical audits, native statuses and timing are retained per decision.

Across both MIQP arms, 0 native bounds exceed the best audited cross-arm objective by more than the unchanged $10^{-12}$ variance comparison scale. Native bounds remain diagnostics, not independently certified bounds. Independent ExactBase is lower on 0 decisions; its maximum delivered advantage is 0.000043 basis points. These separately started outcomes remain visible even when ExactRanked stays within the descriptive equality band. This frozen benchmark tests a distinct public-input cohort, but its low effective candidate counts limit conclusions about dense optional search. The comparison establishes the reported conditional outcomes; it does not establish universal superiority or realised investment gains.

An independent sizing check uses case 0027, with 17 mandatory reservations and no optional candidates. The admissible mandatory state has one movable noncash coordinate. Intersecting its instrument, cash and group intervals and projecting the scalar quadratic's stationary point gives tracking error 79.64086 basis points. Ranked agrees with that independently reconstructed solution to $2.4 \times 10^{-14}$ in maximum absolute weight difference. The alternative fixed state remains infeasible. The check verifies conditional sizing, not global performance on other support patterns.

## 8. Controlled input sensitivity

These retained conditional experiments use only public decision arrays. All treatments preserve the objective form, minimum trades, holding floors

and audit tolerances. Earlier scarce-ticket development results and their complete summaries remain in the replication archive.

*8.1. Input estimates and execution domains*

Twenty-four evenly spaced dates, including both endpoints of the 219-case manifest, receive target-only, covariance-only and joint shocks. Target shocks have 10/50 bp half-$L^1$ NAV distance, preserving cash and zero targets. Covariance shocks use $D\Sigma D$, $D_{ii} = \exp(\epsilon u_i)$, $u_i \in [-1, 1]$, $\epsilon = 0.01/0.05$, preserving positive semidefiniteness. Each runs with the original domain fixed and with optional target corridors rebuilt within unchanged full bounds. Lifecycle declarations, eligibility, group rows and cash limits stay fixed. Common preparation is repeated for rebuilt corridors. This is conditional sensitivity, not upstream estimation, changed lifecycle policy or an own-book return simulation.

All 24 baseline and 288 perturbed portfolios pass the independent audit. Risk changes are also evaluated against the original target and covariance. No case or failed output is removed.

Table 5: Large-shock allocation changes: half-$L^1$ percent of NAV.

| Input | Domain | 95th pct. | Maximum | Support changed |
|---|---|---|---|---|
| Target | Fixed | 0.736 | 0.861 | 0/24 |
| Target | Rebuilt | 0.803 | 0.861 | 5/24 |
| Covariance | Fixed | 0.091 | 0.206 | 0/24 |
| Covariance | Rebuilt | 0.091 | 0.206 | 0/24 |
| Joint | Fixed | 0.669 | 0.932 | 0/24 |
| Joint | Rebuilt | 0.736 | 0.932 | 5/24 |

These dates overlap across treatments; they are not independent replications. The numerical archive retains turnover, common-reference risk and full weights.

## 9. Full joint mixed-integer benchmark

A frozen public synthetic grid crosses 24/72/144 nominal optional instruments, 2/6 discretionary tickets, disjoint/overlapping groups and tight/wide cash and group bands. Seed 111001 supplies 24 paired instances; seed 112001

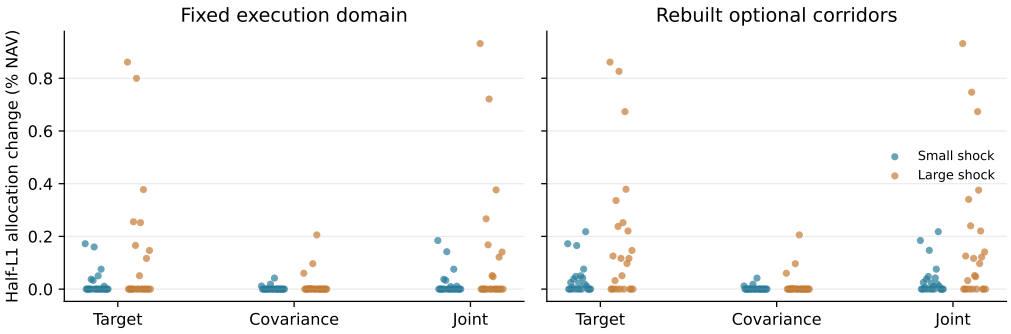



Figure 1: All conditional allocation shifts. Small/large target shocks are 10/50 NAV bp; volatility shocks have maximum absolute log shifts of 1/5 percent.

supplies two eight-instrument controls. The generator retains two forced exits, a 50 NAV bp minimum ticket and a 25 NAV bp holding floor. The V2 heuristic and rules remain frozen. This paired rerun corrects the MIQP reference's native feasibility setting and quadratic-cone scaling after this grid exposed a false optimal termination. It is a numerical regression comparison, not an untouched validation of the corrected reference. Factorial variants share random draws and are not independent market replications.

Each joint MIQP optimises all admissible states and sizes simultaneously in the same prepared CORRIDOR domain. MIP-B starts from the common prepared incumbent; MIP-V receives only V2's audited proposal. Suffixes denote 10/60/600-second wrapper allowances. Every call starts afresh: incumbents never pass between budgets. Native calls use one thread, seed zero and the same $10^{-4}$ relative-gap setting; native integer feasibility is $10^{-9}$. Budget and start orders alternate deterministically. The mathematical MIQP is solved in an algebraically equivalent, balanced conic representation.

All 24 broad cases were attempted; 24 prepared. Effective optional candidates range from 20 to 41. Table 6 reports audited outputs over the full expected denominator. Tracking-error means use audited outputs; time summaries include every attempted arm. End-to-end (E2E) adds preparation to every arm and V2 generation to each MIP-V call.

V2 exceeds the best audited delivery by more than the $10^{-4}$ bp descriptive band on 0/24 cases; its largest delivered gap is 0.000000 bp. A worse time-limited MIP delivery does not imply a worse mathematical optimum. A tie with MIP-V may simply retain V2's supplied incumbent. All

Table 6: Full joint optimisation versus V2. TE is annual bp; E2E time is seconds.

| Arm | Audited | Mean TE | Median E2E | 95th pct. | V2 worse |
|---|---|---|---|---|---|
| V2 | 24/24 | 132.960 | 1.559 | 6.849 | – |
| MIP-B10 | 24/24 | 132.960 | 0.171 | 0.551 | 0 |
| MIP-V10 | 24/24 | 132.960 | 1.892 | 7.202 | 0 |
| MIP-B60 | 24/24 | 132.960 | 0.166 | 0.551 | 0 |
| MIP-V60 | 24/24 | 132.960 | 1.895 | 7.184 | 0 |
| MIP-B600 | 24/24 | 132.960 | 0.165 | 0.555 | 0 |
| MIP-V600 | 24/24 | 132.960 | 1.889 | 7.219 | 0 |

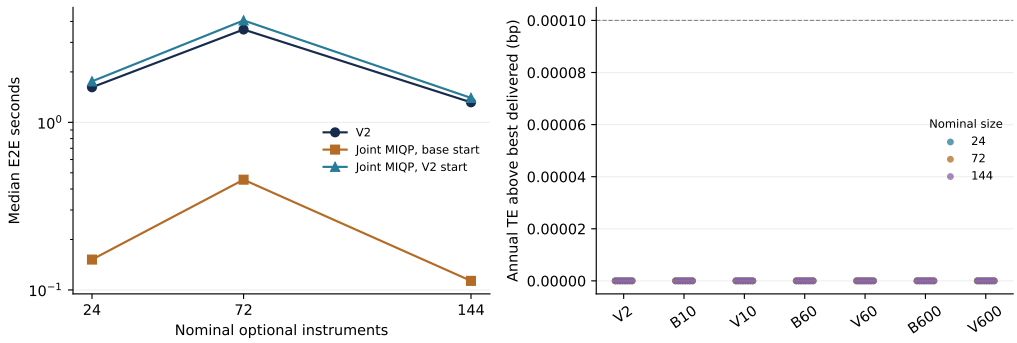



Figure 2: Left: median E2E computation at the 600-second wrapper allowance, including preparation and V2 start generation. Right: each audited outcome above the best delivered TE on that case; this pooled reporting reference is never supplied as a start. Points identify nominal optional-universe size; the dashed line is the unchanged descriptive equality band.

native/retained/refined sources, first native audited feasibility times, failed outputs and allowance overruns are preserved in the numerical summary.

Changing the allowance changes delivered TE above the descriptive band on 0 common-base and 0 V2-started cases. These allowances are upper limits, not enforced run durations. At 600 seconds, V2 has lower TE than the common-base delivery on 0/24 cases and ties on 24/24; its largest advantage is 0.000000 bp. MIP-V600 retains the supplied V2 start on 12 cases.

Across all six MIQP arms, 0 native bounds exceed a better audited cross-arm objective at the unchanged $10^{-12}$ variance comparison scale. Raw bounds and statuses are retained, not clipped to the incumbent. Consistency with delivered witnesses is necessary but does not independently certify the native branch-and-bound result.

Separate conservative relaxation bounds close within the descriptive band on 0/24 cases. Their median/maximum best-delivery gaps are 4.153/20.025 bp; there are 0 upper/lower inconsistencies. Native optimal status and solver-reported MIP gaps remain diagnostics, not independent global certificates.

Independent full-weight QP enumeration covers 74 supports across 2 tiny cases; 2 enumerations are complete and 0 supports remain unresolved. The maximum absolute delivered difference across all tiny arms is 3.55e-09 bp. These are floating-point numerical references, not exact-arithmetic proofs.

The earlier development, holdout and search-ablation records remain in the replication package. This rerun changes only the reference's internal numerical setting, conic scaling and diagnostic checks, preserving the heuristic and production policy. Computational cost is measured rather than assumed: the experiment establishes neither a universal speed advantage nor realised investment performance.

## 10. Limitations and conclusion

Ranked Execution separates cash-funded prioritisation from the joint sizing needed for coupled constraints. Executable states encode minimum trades and holding alternatives; mandatory declarations make counting and exemptions explicit. Coordinated search broadens the combinations considered under a fixed domain, and original-unit audits control delivery.

A better objective than a time-limited mixed-integer call is not a global certificate. An Exact improvement identifies a search or numerical limit that must remain visible. Retained starts, native discoveries and numerical refinements are different sources of delivered portfolios.

The empirical limits include the retrospective universe, supplied rather than refitted estimates, dependent monthly cases and the number of genuinely optional states after preparation. A large nominal universe does not by itself establish difficult sparse search. The retained candidate counts measure the scale actually tested. Numerical corrections were informed by these instances; the sample is not an untouched validation cohort.

This benchmark does not simulate every algorithm's recursively evolving holdings. Such a study requires asset returns and a separate implementation convention; its outcomes combine local decisions with path dependence. No realised return, Sharpe ratio or trading-cost advantage follows from the conditional comparison.

The earlier synthetic studies informed search corrections and remain in the replication archive. The paired rerun preserves V2 and its acceptance rules; the corrected reference was informed by these cases. The grid has 20–41 effective optional candidates, despite nominal sizes up to 144. V2 ties all 24 independent joint-MIQP deliveries within the descriptive band at every allowance. The previously reported 3.645 bp advantage disappears. Independent MIQP has lower median E2E cost (0.165 versus 1.559 seconds at the 600-second allowance). Heuristic-started MIQP matches V2 but must also pay its generation cost. None of these calls uses the full allowance; longer allowances provide no measured improvement.

The corrected ETF and synthetic comparisons have no native bound contradicting an audited cross-arm witness by more than $10^{-12}$ variance. Raw bounds remain visible; this consistency check is not an independent optimality certificate. Conservative relaxation bounds remain valid but do not close the broader cases. The tiny enumerations agree with all arms within the descriptive band. Together these controls support neither universal dominance nor a heuristic speed advantage, and they do not certify global optimality on the larger cases.

The controlled sensitivity study also shows that changing input estimates and changing the execution domain are distinct effects. All studies remain conditional on supplied inputs and finite search budgets. External validation, broader lifecycle changes, rolling own-book simulations and matched established alternatives remain needed. Further algorithm changes would require fresh qualification; the present results do not authorize a production change. The contribution is an auditable execution heuristic with explicit controls and reproducible evidence of both improvements and limits.

## Data and code availability

The accompanying package contains de-identified numerical decision inputs, checksums, the ETF and synthetic runners, independent audits, reference enumeration and table-generation code. The reusable solver is in OptimalPortfolios. Tracking-error reporting uses QIS. This public-facing draft is prepared for release; a persistent dataset identifier and public release URL have not yet been assigned. Execution conditional on supplied estimates is reproduced; upstream allocation-model estimation and raw market-data histories are outside the release.

## Author declarations

Affiliation, competing-interest and final authorship statements require author completion before submission. AI assistance was used in implementation, analysis and drafting; the author must review and approve the final disclosure.

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